

Multi-agent Deep Reinforcement Learning

Learning with multiple agents

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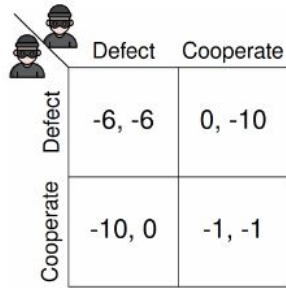
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- Definitions
- Learning with multiple agents
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- MADRL research

Introduction

- Multi-agent systems
- From single to multi-agent systems
- Multi-agent deep reinforcement learning

Introduction

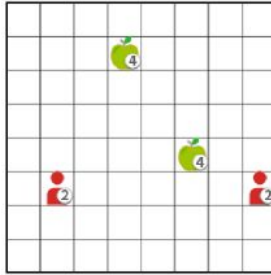
Multi-agent system



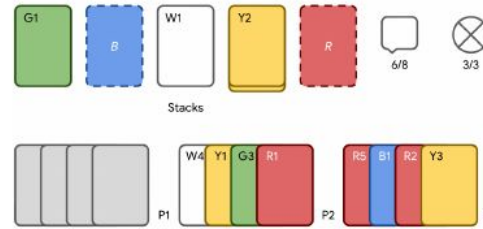
A payoff matrix for a social dilemma game. The rows and columns are labeled 'Defect' and 'Cooperate' with corresponding robot icons. The payoffs are as follows:

	Defect	Cooperate
Defect	-6, -6	0, -10
Cooperate	-10, 0	-1, -1

(a) Social dilemma



(b) Two-dimensional grid world



(c) Card game



(d) Multi-player video game



(e) Multi-robot system



(f) Human-robot interaction

Introduction

From single to multi-agent systems

Agents Computer system that interacts with its environment, considered intelligent for being:

- ***reactive***: perceives information and acts in response to perceived signals
- ***proactive***: shapes its behaviour to satisfy its objectives
- ***learning***: learns from its experience

Multi-agent systems A system of multiple intelligent agents that interact with their environment and with each other.

-> Agents are also:

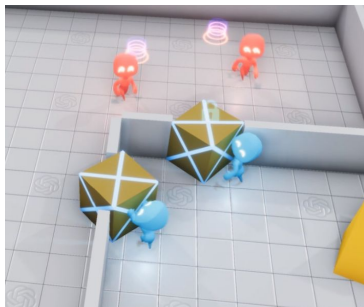
- ***social***: shapes its behaviour depending on the behaviour of other agents

Introduction

Multi-agent deep reinforcement learning



OpenAI Five (2019)⁽¹⁾



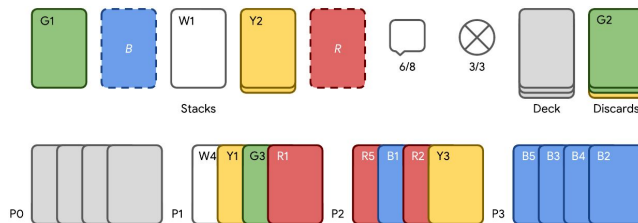
Hide-and-seek (2019)⁽²⁾



Google Research Football (2019)⁽⁵⁾



Starcraft Multi-Agent Challenge (2019)⁽³⁾



Hanabi (2019)⁽⁴⁾

⁽¹⁾OpenAI et al., *Dota 2 with Large Scale Deep Reinforcement Learning*, 2019

⁽²⁾Baker et al., *Emergent Tool Use From Multi-Agent Autocurricula*, 2019

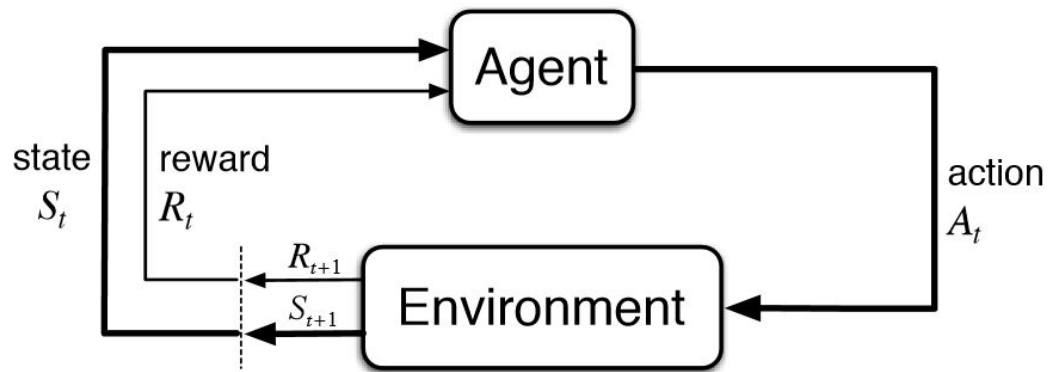
⁽³⁾Samvelyan et al., *The StarCraft Multi-Agent Challenge*, 2019

⁽⁴⁾Bard et al., *The Hanabi challenge: A new frontier for AI research*, 2020

⁽⁵⁾Kurach et al., *Google Research Football: A Novel Reinforcement Learning Environment*, 2019

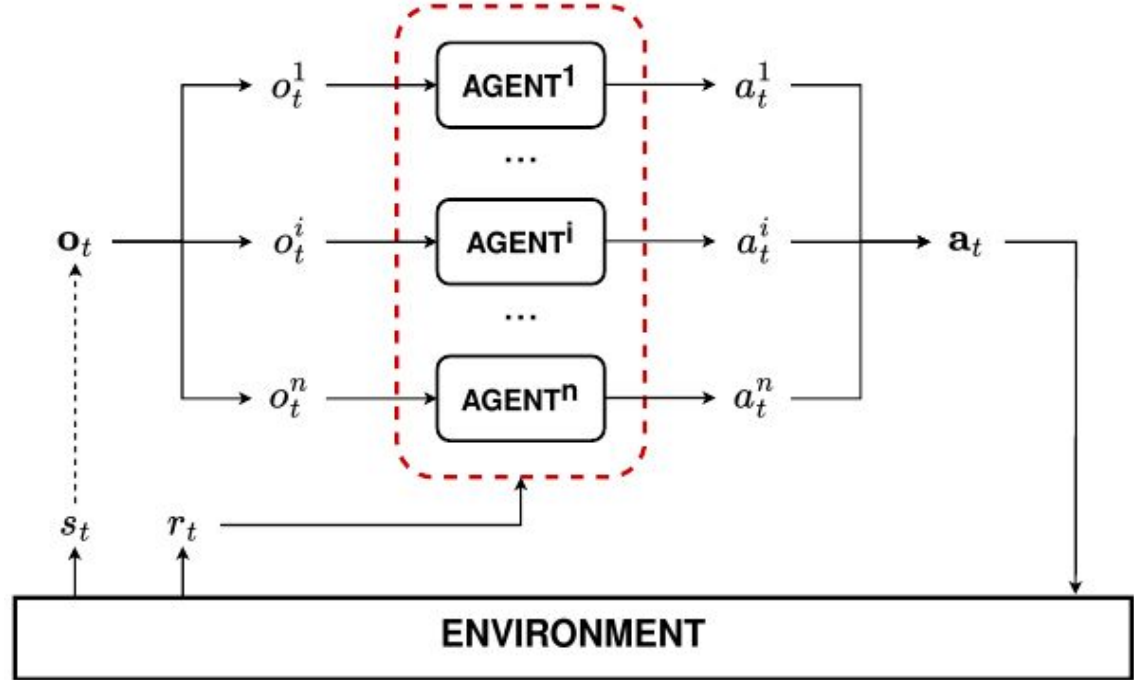
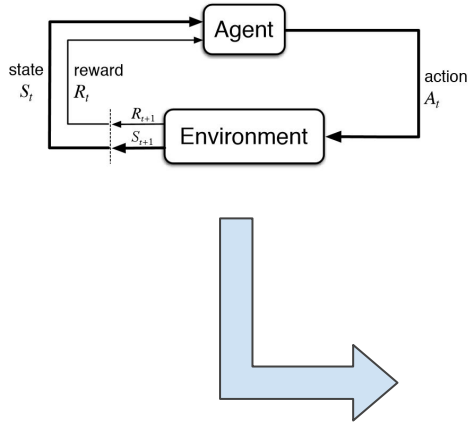
Introduction

Multi-agent deep reinforcement learning



Introduction

Multi-agent deep reinforcement learning



Definitions

- Dec-POMDP
- Multi-agent RL tools
- Nash equilibrium
- Pareto optimality

Definition

Decentralised-Partially Observable MDP

$$\langle \mathbf{S}, \mathbf{A}, \mathcal{T}, \mathbf{O}, \mathcal{O}, \mathcal{R}, n, \gamma \rangle$$

n number of agents

$\mathbf{S}$ environment states, “global states”

$\mathbf{O}$ set of **joint observations** $\mathbf{o} = \{o^1, \dots, o^n\}$

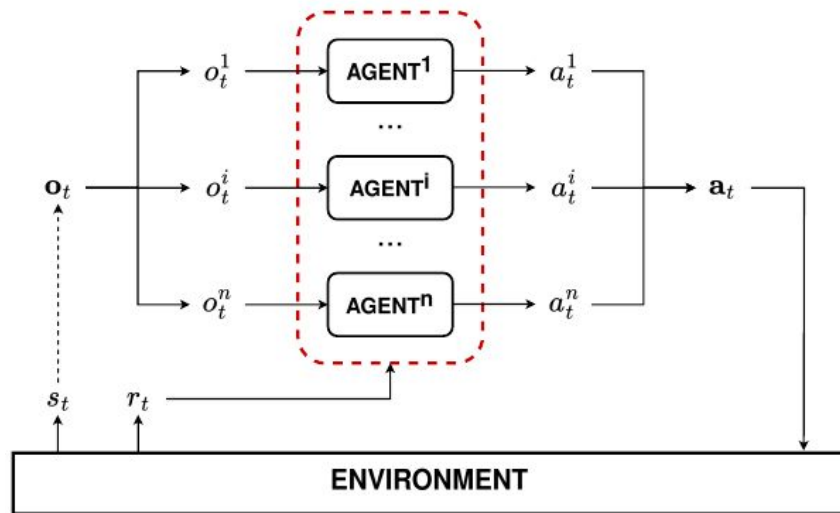
$\mathbf{A}$ set of **joint actions** $\mathbf{a} = \{a^1, \dots, a^n\}$

$\mathcal{T}$ transition function $P(s'|s, \mathbf{a})$

$\mathcal{O}$ observation function $P(\mathbf{o}|\mathbf{a}, s')$

$\mathcal{R}$ reward function

γ discount factor



Definition

Multi-agent RL tools: joint and local policies

Joint Policy $\pi = (\pi^1, \dots, \pi^n)$

Local policies $\pi^i(a^i|h^i) : \mathbf{H} \times \mathbf{A} \rightarrow [0, 1]$

history $h_t^i = (o_0^i, a_0^i, \dots, a_{t-1}^i, o_t^i) \in \mathbf{H} = (\mathbf{O} \times \mathbf{A})^*$

Definition

Multi-agent RL tools: joint and local values

Joint value functions

$$V_{\pi}(\mathbf{h}_t) = \mathbb{E}_{\pi}[G_t \mid \mathbf{h}_t]$$

$$Q_{\pi}(\mathbf{h}_t, \mathbf{a}_t) = \mathbb{E}_{\pi}[G_t \mid \mathbf{h}_t, \mathbf{a}_t]$$

Local value functions

$$V_{\pi}(h_t^i) = \mathbb{E}_{\pi}[G_t \mid h_t^i]$$

$$Q_{\pi}(h_t^i, a_t^i) = \mathbb{E}_{\pi}[G_t \mid h_t^i, a_t^i]$$

Definition

Nash Equilibrium

A **Nash equilibrium** is a situation where no player could gain by changing their own strategy, *without harming the other players' gain*.

- Not necessarily unique

Pure-strategy equilibrium
→ deterministic joint policy

(A,A) or (B,B)

	A	B
A	1,1	1,-1
B	-1,1	0,0

Definition

Nash Equilibrium

A **Nash equilibrium** is a situation where no player could gain by changing their own strategy, *without harming the other players' gain*.

Mixed equilibrium
→ stochastic joint policy

$p(r)=\frac{1}{3}$ $p(p)=\frac{1}{3}$ $p(s)=\frac{1}{3}$
for both players

1 \ 2	r	p	s
R	(0, 0)	(-1, 1)	(1, -1)
P	(1, -1)	(0, 0)	(-1, 1)
S	(-1, 1)	(1, -1)	(0, 0)

Definition

Pareto optimality

An outcome of a game is **Pareto optimal** if there is no other outcome that makes every player at least as well off and at least one player strictly better off.

(A, A)

	A	B
A	1,1	1,-1
B	-1,1	0,0

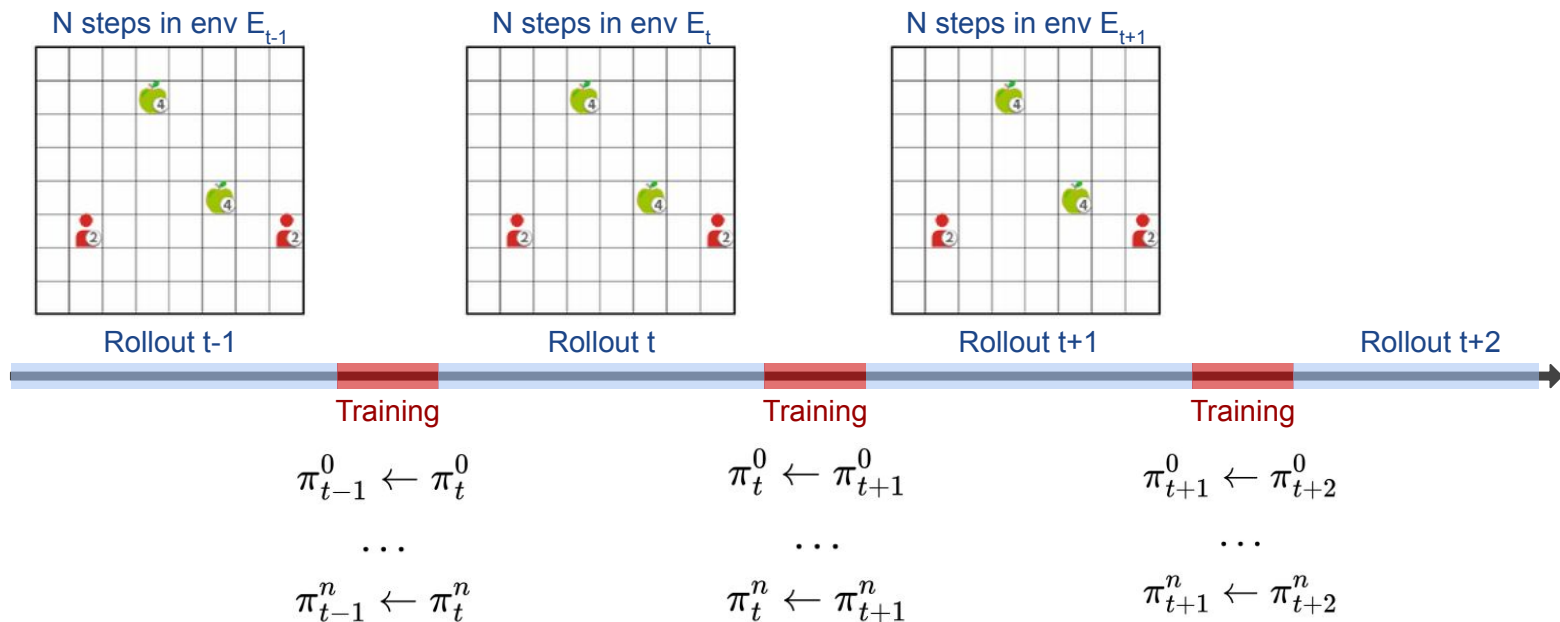
Learning with multiple agents

- Non-stationarity
- Partial observability
- Credit assignment
- Scaling

Learning with multiple agents

Non-stationarity

From a single agent's point of view, other agents are part of the environment.
→ If other agents change their strategy, the environment changes



Learning with multiple agents

Non-stationarity

From a single agent's point of view, other agents are part of the environment.

→ If other agents change their strategy, the environment changes

Problem: Experience replay is broken (Foerster2016)

Experience from old steps are not relevant because the agents have changed too much

→ **Concurrent experience replay** (Omidshafiei2017)

Train all agents on the same environment steps, i.e., similar data will help agents learn similar things

→ **Weight down old experiences** (Foerster2017)

→ **Use smaller replay memory**

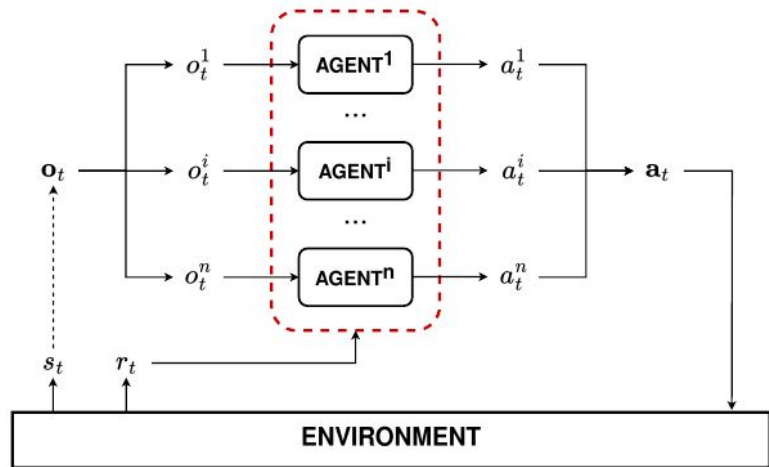
Learning with multiple agents

Partial observability

Agents gather only **incomplete information** about the environment, i.e., the local observation resulting from sensors.

$$\mathcal{O}(\mathbf{a}, s') = P(\mathbf{o}|\mathbf{a}, s')$$
$$\mathbf{o} = \{o^1, \dots, o^n\}$$

Each agent has a different local observation...
→ agents have asymmetrical knowledge of the environment



Learning with multiple agents

Partial observability

Agents gather only **incomplete information about the environment**, i.e., the local observation resulting from sensors.

Problem: Agents need to behave according to incomplete knowledge of the current state

- **Beliefs?** difficult as the state depends on other agents (Oliehoek2016)
- **Memory** (Hausknecht2015)
Memorise previously gathered information for better estimating the current state
- **Communication** (Zhu2024)
Share local information with other agents to cooperatively build better state estimation

Learning with multiple agents

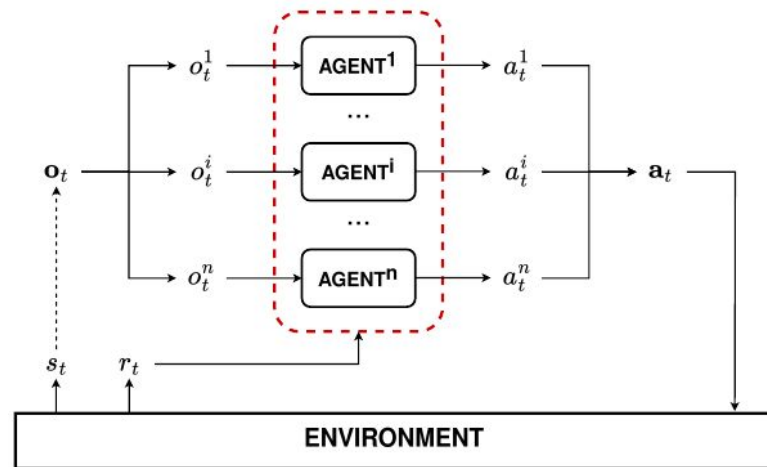
Credit assignment

“Did I succeed (or fail) because of my actions, or because of the actions of other agents?”

i.e.,

How to measure the contribution of each agent’s actions?

$$\mathcal{R} : \mathbf{S} \times \mathbf{A} \rightarrow \mathbb{R},$$
$$\mathcal{R}(s_t, \mathbf{a}_t) = r_{t+1}$$



Learning with multiple agents

Credit assignment

Credit assignment aims to find the real **marginal contribution** of each action.

$$r_{t+1} = \sum_i u_{t+1}^i, \text{ with } u_{t+1}^i = u(a_t)$$

→ **Shapley value** (Shapley1953) $u^i = \mathbb{E}_{C \subset N \setminus \{i\}} [u(C \cup i) - u(C)]$

i.e., contribution to any possible coalition

→ **Wonderful Life Utility** (Wolpert1999) $u^i = u(\mathbf{a}) - u(\mathbf{a}_{-i}, a_i = \text{null})$

i.e., contribution compared to counterfactual baseline

→ **Aristocrat Utility** (Wolpert2002) $u^i = u(\mathbf{a}) - \mathbb{E}[u | \mathbf{a}_{-i}, s]$

Learning with multiple agents

Scaling

*How to learn in larger environments, **with more agents** ?*

Problems:

- worse non-stationarity
- harder credit assignment
- harder joint policy learning

→ **Deep learning?**

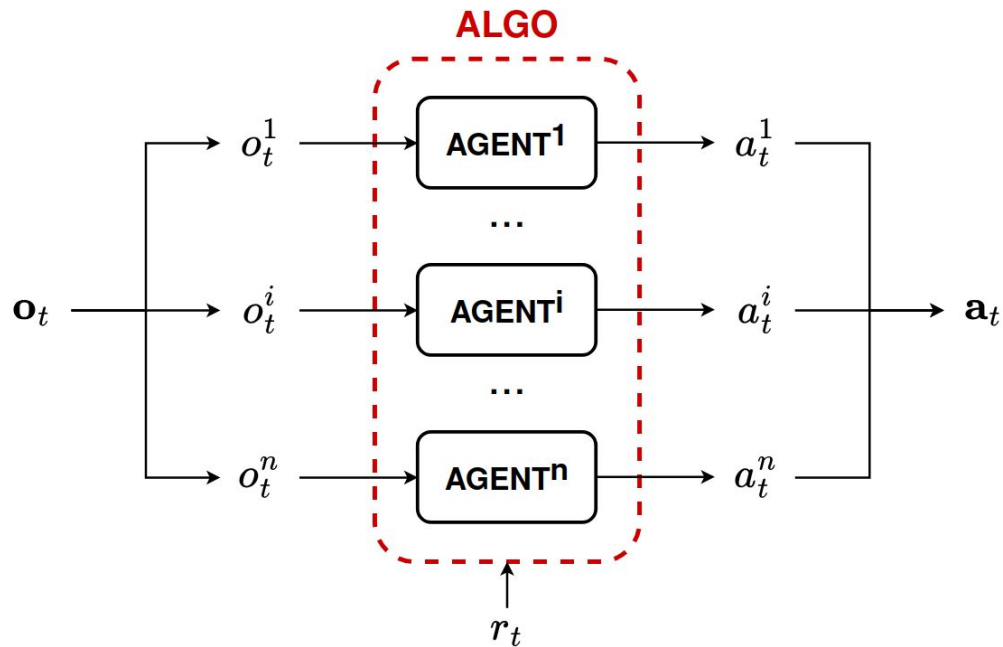
Multi-agent deep reinforcement learning

- Multi-agent learning paradigms
- Independent learning
- Multi-agent actor-critics
- Value factorisation
- Emergent communication
- Agent modelling

Multi-agent (deep) reinforcement learning

Learning paradigms

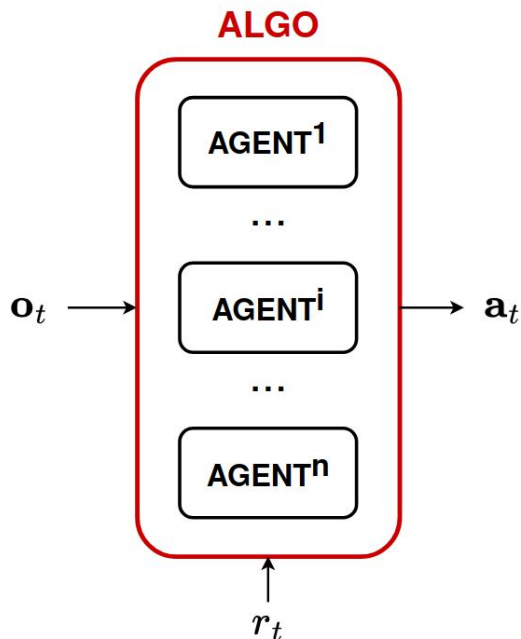
How to define the learning algorithm?



Multi-agent (deep) reinforcement learning

Learning paradigms: Reducing to single-agent RL

Fully centralised

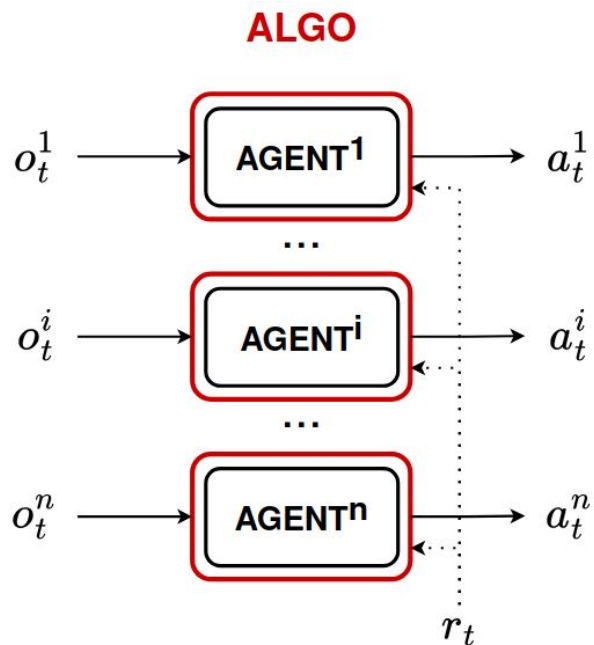


- + No more non-stationarity
- + No more credit assignment
- + Better partial observability
- Worse scaling ability:
 - larger observation space
 - larger action space
- Not possible in realistic environments

Multi-agent (deep) reinforcement learning

Learning paradigms: Reducing to single-agent RL

Fully decentralised

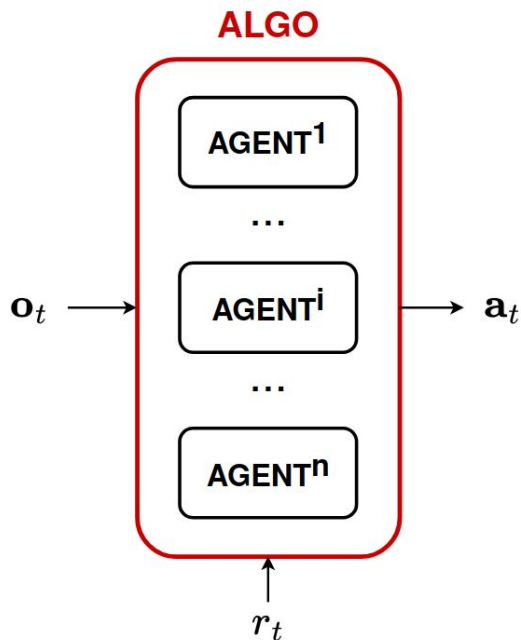


- + Better scaling
- + More compute-efficient
- Non-stationarity
- Credit assignment

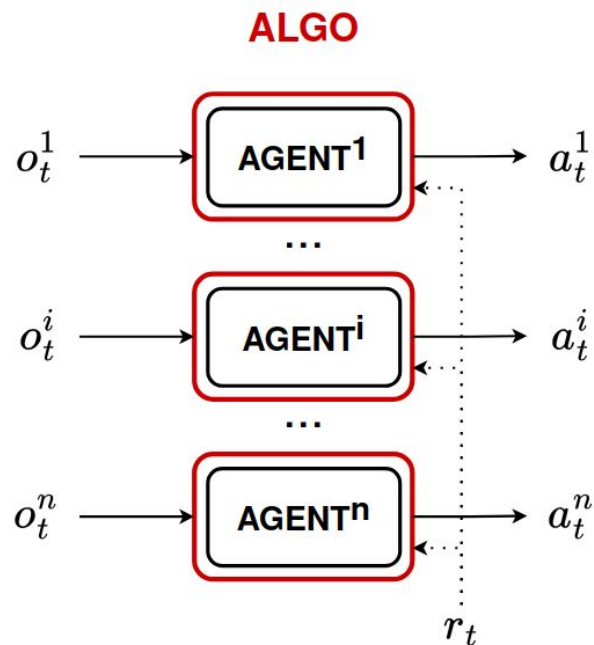
Multi-agent (deep) reinforcement learning

Learning paradigms: Reducing to single-agent RL

Fully centralised



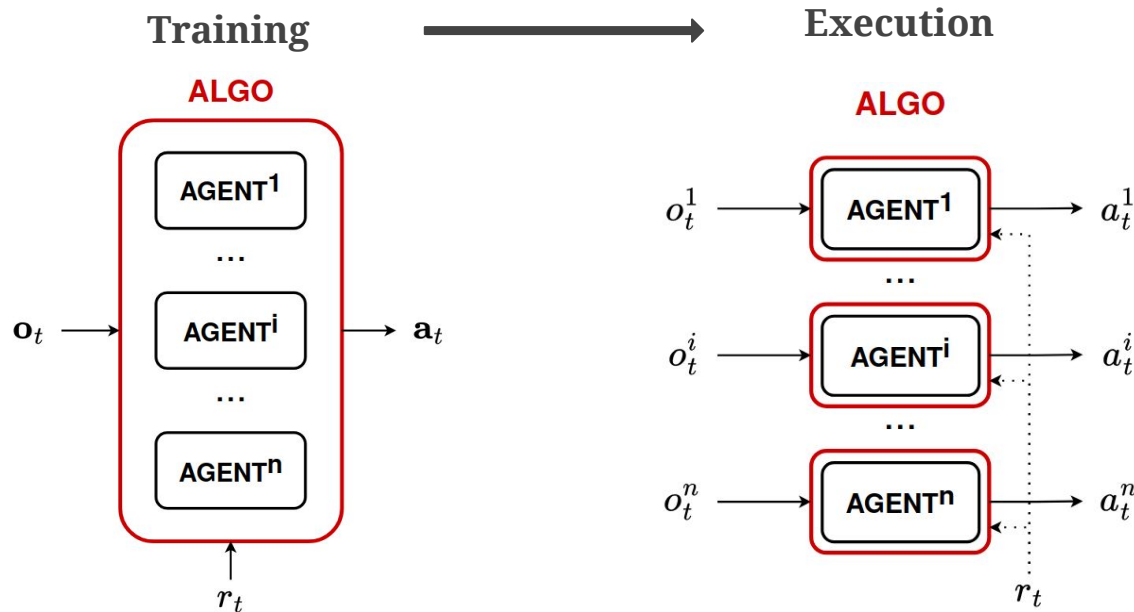
Fully decentralised



Multi-agent (deep) reinforcement learning

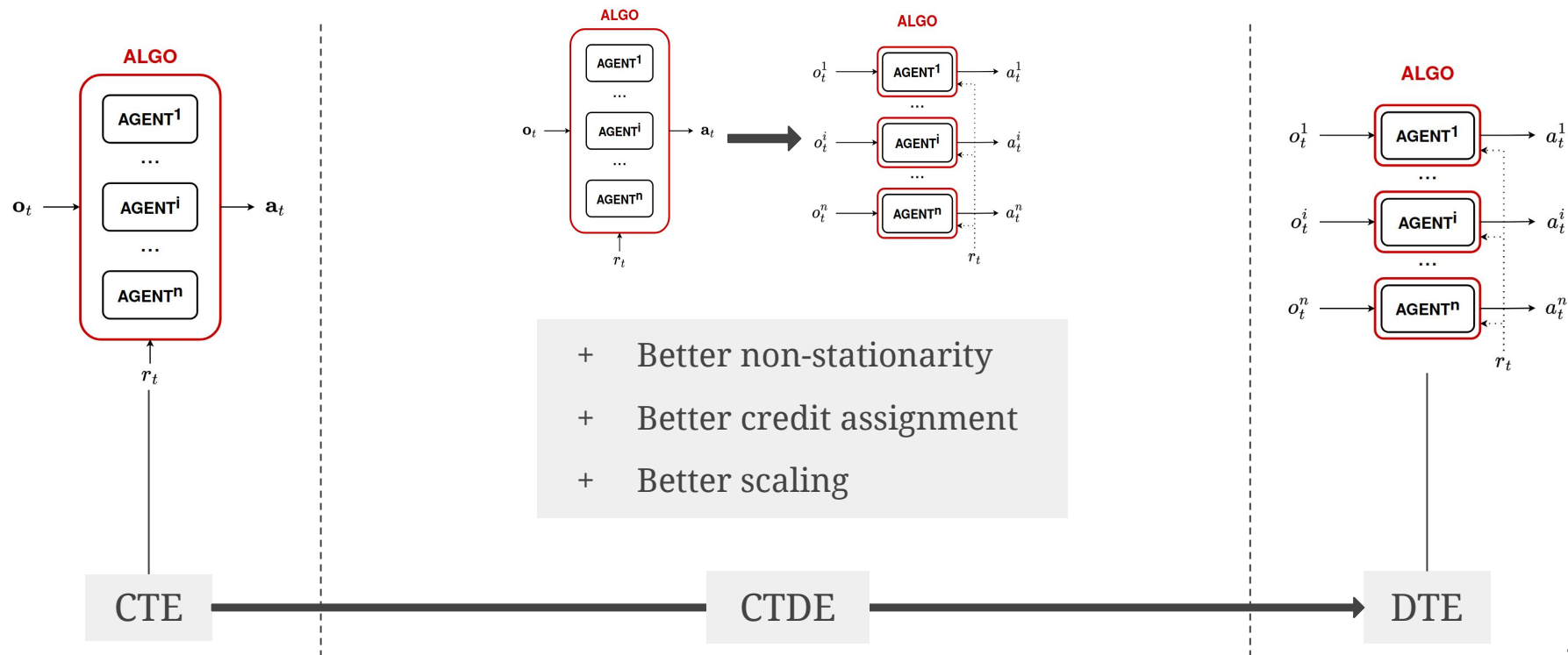
Learning paradigms: Training/Execution

Centralised Training with Decentralised Execution



Multi-agent (deep) reinforcement learning

Learning paradigms: Centralised Training with Decentralised Execution



Multi-agent (deep) reinforcement learning

Independent Learning

Independent Learning: Considering each agent independently... with some simplifications. (see Jiang2024)

- **Independent Q-learning** (Tan1993)
 - sharing parameters between agents
- **Concurrent experience replay** (Omidshafiei2017)
- **Population-based training** (Jaderberg2019)
 - training large number of agents, with changing teams → better generalisation
- **Ideal Independent Q-learning** (Jiang2022)
 - Consider other agents will act optimally

Tan, *Multi-Agent Reinforcement Learning: Independent versus Cooperative Agents*, 1993

Omidshafiei et al., *Deep Decentralized Multi-task Multi-Agent Reinforcement Learning under Partial Observability*, 2017

Jaderberg et al., *Human-level performance in first-person multiplayer games with population-based deep reinforcement learning*, 2019

Jiang and Lu, *I2Q: A Fully Decentralized Q-Learning Algorithm*, 2022

Jiang et al., *Fully Decentralized Cooperative Multi-Agent Reinforcement Learning: A Survey*, 2024

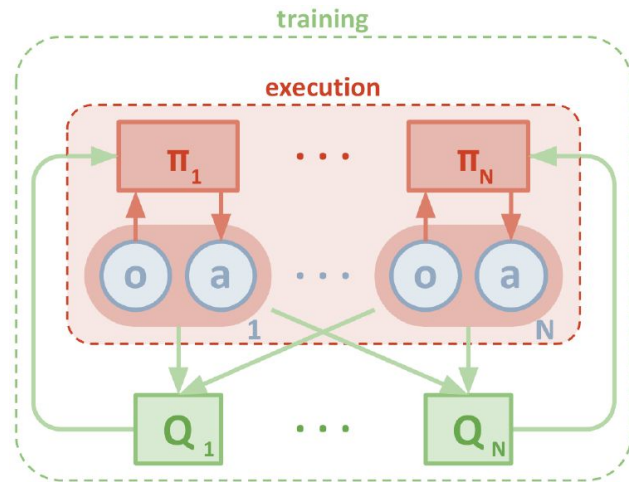
Multi-agent (deep) reinforcement learning

Multi-agent actor-critics

To fit the CTDE paradigm, agents only need to have decentralised policies.
→ value function can be centralised, as they're used only during training

→ **Multi-Agent DDPG** (Lowe2017)

- decentralised policies, trained by centralised critics
- one policy per agent $\pi_i(o^i) = a^i$
- one critic per agent $Q_i(\mathbf{o}, a^i)$



Multi-agent (deep) reinforcement learning

Multi-agent actor-critics

To fit the CTDE paradigm, agents only need to have decentralised policies.
→ value function can be centralised, as they're used only during training

- **Multi-Agent TD3** (Ackermann2019)
- **Multi-Agent SAC** (Yu2021a)
- **Multi-Agent PPO** (Yu2021b)
- **COMA** (Foerster2018)
 - Aristocrat utility computed with a centralised critic:

$$A^i(\mathbf{h}, \mathbf{a}) = Q(\mathbf{h}, \mathbf{a}) - \sum_{a^{i'}} \pi^i(a^{i'} | h^i) Q(\mathbf{h}, (\mathbf{a}^{-i}, a^{i'}))$$

Multi-agent (deep) reinforcement learning

Value factorisation

How can we learn the value of local actions?

$$Q_{\pi}(h_t^i, a_t^i) = \mathbb{E}_{\pi}[G_t \mid h_t^i, a_t^i]$$

Problem: This value uses the return G_t , but G_t depends on the actions of other agents...

To use Q-learning for action-selection, we may instead need:

$$Q_{\pi^i}(h^i, a^i) = \mathbb{E}_{\pi^i} [U_t^i \mid h_t^i, a_t^i],$$

$$\text{with, } U_t^i = \sum_{k=t}^T \gamma^{k-t} u_k^i$$

Multi-agent (deep) reinforcement learning

Value factorisation

How can we learn the value of local actions?

$$Q_{\pi^i}(h^i, a^i) = \mathbb{E}_{\pi^i} [U_t^i | h_t^i, a_t^i]$$

This is a **credit assignment issue**, i.e., we need to find how these local values compose the global one that estimates the return:

$$Q(\mathbf{h}, \mathbf{a}) = f(\{Q^i(h^i, a^i)\}_{1 \leq i \leq n}, \mathbf{h})$$

$$Q_{\pi}(\mathbf{h}_t, \mathbf{a}_t) = \mathbb{E}_{\pi}[G_t | \mathbf{h}_t, \mathbf{a}_t]$$

Multi-agent (deep) reinforcement learning

Value factorisation

*How can we compose local action-value into the global one?
i.e., we need to find a function f*

$$Q(\mathbf{h}, \mathbf{a}) = f(\{Q^i(h^i, a^i)\}_{1 \leq i \leq n}, \mathbf{h})$$

$$Q_{\pi^i}(h^i, a^i) = \mathbb{E}_{\pi^i} [U_t^i | h_t^i, a_t^i] \quad Q_{\pi}(\mathbf{h}_t, \mathbf{a}_t) = \mathbb{E}_{\pi} [G_t | \mathbf{h}_t, \mathbf{a}_t]$$

→ To select local actions with Q^i , we need to guarantee the **Individual-Global-Max property**:

$$\operatorname{argmax}_{\mathbf{a}} Q(\mathbf{h}, \mathbf{a}) = \{\operatorname{argmax}_{a^i} Q^i(h^i, a^i)\}_{1 \leq i \leq n}$$

→ function f must guarantee the IGM

Multi-agent (deep) reinforcement learning

Value factorisation

How can we compose local action-value into the global one?
i.e., we need to find a function f **that guarantees IGM**

$$\text{IGM: } \operatorname{argmax}_{\mathbf{a}} Q(\mathbf{h}, \mathbf{a}) = \{\operatorname{argmax}_{a^i} Q^i(h^i, a^i)\}_{1 \leq i \leq n}$$

→ **Value Decomposition Networks (VDN)** (Sunehag2018)

$$Q(\mathbf{h}, \mathbf{a}) \approx \sum_{i=1}^n Q^i(h^i, a^i)$$

- One DQN Q^i for each agent
- All local DQNs learnt from the MSBE of the global Q :

$$L(\theta) = \mathbb{E} \left[\left(r + \gamma \max_{\mathbf{a}'} Q(\mathbf{h}', \mathbf{a}') - Q(\mathbf{h}, \mathbf{a}) \right)^2 \right]$$

Multi-agent (deep) reinforcement learning

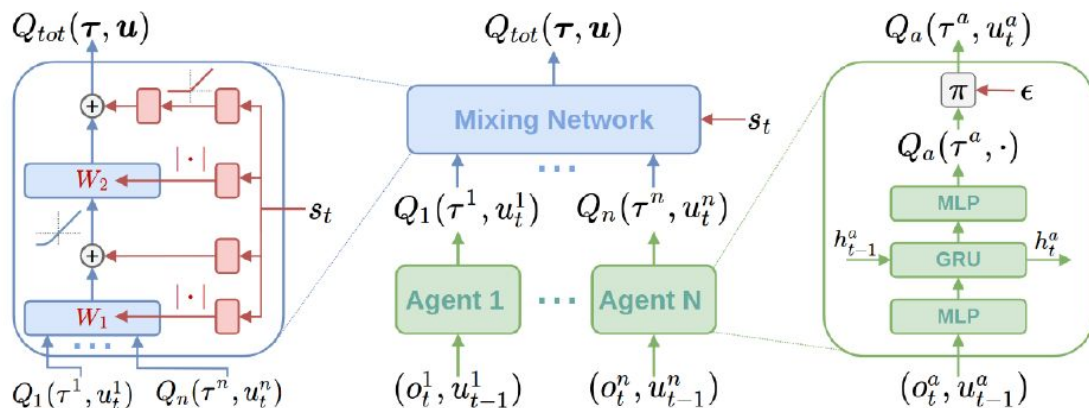
Value factorisation

How can we compose local action-value into the global one?
i.e., we need to find a function f **that guarantees IGM**

$$\text{IGM: } \operatorname{argmax}_{\mathbf{a}} Q(\mathbf{h}, \mathbf{a}) = \{\operatorname{argmax}_{a^i} Q^i(h^i, a^i)\}_{1 \leq i \leq n}$$

→ **QMIX** (Rashid2018)

- f modeled as neural network
- f takes the state as input, i.e., **the mixing strategy depends on the state**
- f 's weights are made positive to ensure the IGM property



Multi-agent (deep) reinforcement learning

Value factorisation: QMIX extension

- **VDN** (Sunehag2018)
- **QMIX** (Rashid2018)
- **QTRAN** (Son2019)
- **Weighted QMIX** (Rashid2020)
- **QPLEX** (Wang2021): IGM based on Advantage function instead of Q-value
- **Qatten** (Yang2020): attention in mixing network
- **LICA** (Zhou2020) & **FACMAC** (Peng2021): QMIX in actor-critic framework
- **DFAC** (Sun2023): QMIX adapted to distributional DQN

Sunehag et al., *Value-Decomposition Networks For Cooperative Multi-Agent Learning Based On Team Reward*, 2018

Rashid et al., *QMIX: Monotonic Value Function Factorisation for Deep Multi-Agent Reinforcement Learning*, 2018

Son et al., *QTRAN: Learning to Factorize with Transformation for Cooperative Multi-Agent Reinforcement Learning*, 2019

Rashid et al., *Weighted QMIX: Expanding Monotonic Value Function Factorisation for Deep MultiAgent Reinforcement Learning*, 2020

Wang et al., *QPLEX: Duplex Dueling Multi-Agent Q-Learning*, 2021

Yang et al., *Qatten: A General Framework for Cooperative Multiagent Reinforcement Learning*, 2020

Zhou et al., *Learning Implicit Credit Assignment for Cooperative Multi-Agent Reinforcement Learning*, 2020

Peng et al., *FACMAC: Factored Multi-Agent Centralised Policy Gradients*, 2021

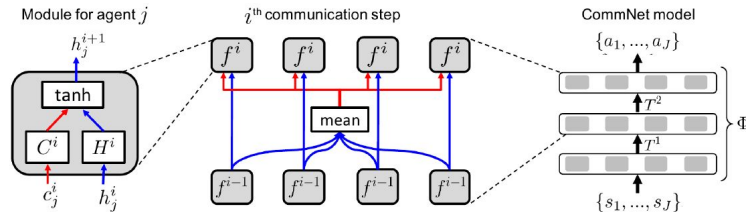
Sun et al., *A Unified Framework for Factorizing Distributional Value Functions for Multi-Agent Reinforcement Learning*, 2023

Multi-agent (deep) reinforcement learning

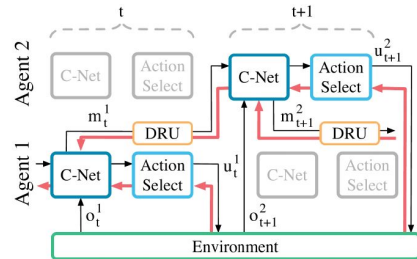
Emergent communication

Agents can learn to communicate to maximise their return, i.e., **emergent communication** (Lazaridou2020)

→ **CommNet** (Sukhbaatar2016)



→ **DIAL** (Foerster2016)



Multi-agent (deep) reinforcement learning

Agent modelling

Agents can learn to predict the other agents' behaviour (Albrecht2018)

- **Fictitious play** (Brown1951, Robinson1951)
 - track actions of other agents to estimate their policy
- **Bayesian learning** (Jordan1991, Foerster2019)
 - track probabilities over possible policies
- + Allows to better adapt to new partners

Albrecht and Stone, *Autonomous agents modelling other agents: A comprehensive survey and open problems*, 2018

Brown, *Iterative solution of games by fictitious play*, 1951

Robinson, *An Iterative Method of Solving a Game*, 1951

Jordan, *Bayesian learning in normal form games*, 1991

Foerster et al., *Bayesian Action Decoder for Deep Multi-Agent Reinforcement Learning*, 2019

MADRL research

- Benchmarking MADRL
- Exploration
- Zero-shot teaming
- Human-robot interaction
- Agent-based LLMs and LLM-societies

MADRL Research

Benchmarking MADRL

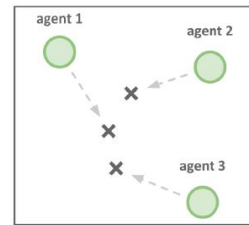
Hard to compare fairly all approaches.

But, at the same time, research runs for best performance

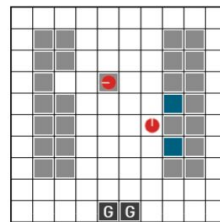
- Many different environments, but not so much diversity
 - Quality of envs. and tasks are not well questioned
 - What skills are we testing?
 - Hard to compare with literature:
 - Many different methods, for many different settings
 - MADRL is expensive to train
 - Numerous hyperparameters (like DRL, but worse)
 - Implementation are complex, with implementation tricks (like DRL, but worse) → discrepancies in results
 - Different programming library used
- Attempts at benchmarking (Papoudakis2021)
- Attempts at instauring standardised evaluation (Gorsane2022)



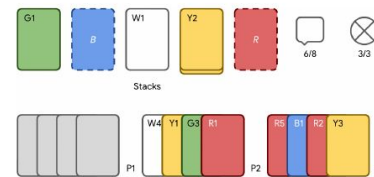
(a) SMAC



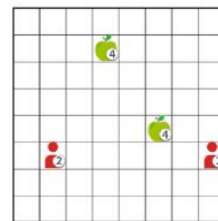
(b) MPE



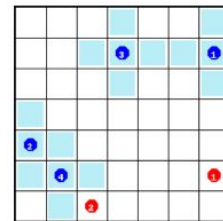
(d) RWARE



(e) Hanabi



(c) LBF

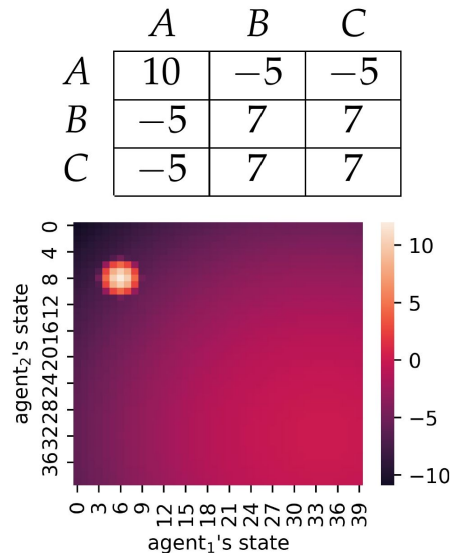


(f) magym

MADRL requires exploration, like RL, but multi-agent exploration requires special care.

Problem: Exploring locally $\neq$ Exploring jointly

i.e., exploring the joint-observation space requires coordinated exploration



Agents should be able to interact with unknown partners (Kirk2023)

Problem: Training a team of agents will results in overfitting on the team collective behaviour

- **Agent modelling** (Albrecht2018)
- **Population-based training** (Jaderberg2019)
- **Predicting the “type” of observed agent** (Lanctot2023)

One goal of MADRL is robot application → interaction is a challenge

Understanding:

- how humans behave (Shih2021)
- how humans interact with each other (Tseng2016)
- how humans react to robots (Roesler2024)
- how humans communicate in social interactions (Feine2019)

Designing:

- physical interfaces (Zlotowski2014)
- social interfaces (Liu2022)

Shih et al., *On the Critical Role of Conventions in Adaptive Human-AI Collaboration*, 2021

Tseng et al., *Service robots: System design for tracking people through data fusion and initiating interaction with the human group by inferring social situations*, 2016

Roesler et al., *The dynamics of human-robot trust attitude and behavior — Exploring the effects of anthropomorphism and type of failure*, 2024

Feine et al., *A Taxonomy of Social Cues for Conversational Agents*, 2019

Zlotowski et al., *Anthropomorphism: Opportunities and Challenges in Human-Robot Interaction*, 2014

Liu et al., *An analysis of children' interaction with an AI chat-bot and its impact on their interest in reading*, 2022

LLMs can be prompted to follow certain behaviours.
Thus, they can be used to simulate multi-agent interactions.

- **Agent-based LLMs** (Guo2024): LLM prompted to act as an agent (human/robot/...)
 - for robotic control (Driess2023)
 - for Human-AI interaction (Liu2024)
- **LLM-society**: Groups of LLM agents
 - for simulating multi-agent interactions (Park2023, Li2023)
 - for simulating cultural evolution (Perez2024)

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Thank you for you attention !

Questions ?